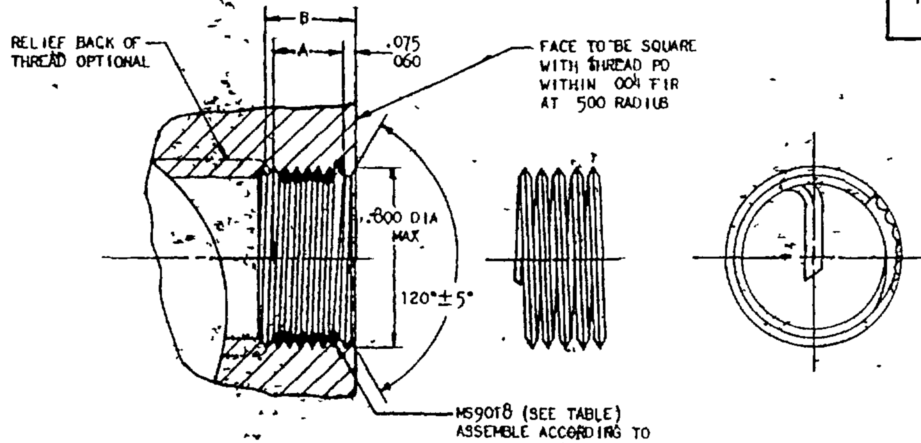
FED SUP CLASS
2810

MS9071 (ASG)


RELIEF BACK OF THREAD OPTIONAL
B
A
.075
.060
FACE TO BE SQUARE WITH THREAD PD WITHIN .004 FIR AT .500 RADIUS
.800 DIA MAX
120°±5°
MS9018 (SEE TABLE) ASSEMBLE ACCORDING TO GENERAL INSTRUCTIONS

ASSEMBLED INSERT

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PART	INSERT PART NUMBER	TAPPED HOLE						ASSEMBLED INSERT				
		PITCH DIA		MAJOR DIA	MINOR DIA		FULL THD DEPTH B	PITCH DIA		MAJOR DIA	ASSEMBLED LENGTH A (REF ONLY)	
		MIN	MAX	MIN	MIN	MAX	MIN	MIN	MAX	MIN	MIN	MAX
LONG REACH												
STANDARD	MS9018-01	.7523	.7543	.793	.718	.723	.772	.6712	.6753	.709	.668	.683
+.003	MS9018-02	.7553	.7573	.796								
+.005	MS9018-03	.7573	.7593	.798								
+.010	MS9018-04	.7623	.7643	.803								
+.015	MS9018-09	.7673	.7693	.808								
+.020	MS9018-10	.7723	.7743	.813								
+.025	MS9018-11	.7773	.7793	.818								
SHORT REACH												
STANDARD	MS9018-05	.7523	.7543	.793	.718	.723	.423	.6712	.6753	.709	.319	.334
+.003	MS9018-06	.7553	.7573	.796								
+.005	MS9018-07	.7573	.7593	.798								
+.010	MS9018-08	.7623	.7643	.803								
+.015	MS9018-12	.7673	.7693	.808								
+.020	MS9018-13	.7723	.7743	.813								
+.025	MS9018-14	.7773	.7793	.818								

GENERAL INSTRUCTIONS

DESCRIPTION: SPARK PLUG THREAD HELICAL COIL INSERTS ARE SCREW THREAD BUSHINGS COILED FROM DIAMOND-SHAPED CROSS SECTION WIRE. THEY ARE SCREWED INTO TAPPED HOLES TO FORM NOMINAL SIZE 18-1.50 MM THREADS. INSERTS ARE ASSEMBLED BY TORQUING THRU A NOTCHED DIAMETRAL TANG. THE TOP OUTER END COIL IS FLAT AND SERRATED AFTER THE INSERT IS IN POSITION THE SERRATED END IS OFF-SET BY AN OFFSET AND STAKING TOOL AND STAKED FIRMLY IN PLACE.

THREADS: ASSEMBLED INSERT THREAD SIZE IS DETERMINED BY TOLERANCE RANGE OF TAPPED HOLE INTO WHICH INSERT IS FITTED. IT IS IMPORTANT WHEN USING OVERSIZE INSERTS THAT AMOUNT OF OVERSIZE ON THE INSERT IS THE SAME AS THAT IN THE TAPPED HOLE, SO THAT THE RESULTING INSIDE THREAD IS OF NOMINAL SIZE.

TAPPED HOLE: 60° AMERICAN STANDARD FORM; TAP SIZES AND GAGES ARE STANDARD WITH MANUFACTURER.

THIS IS A DESIGN STANDARD - NOT TO BE USED AS A PART NUMBER.
DIMENSIONS IN INCHES.

DEPTH OF HOLE: THE PROXIMITY OF THREADS IN TAPPED HOLE AND HELICAL THREAD INSERT TO COMBUSTION CHAMBER TEMPERATURES MAKES IT DESIRABLE TO HAVE AS SMALL AN AREA OF UNUSED THREAD AS PRACTICABLE EXPOSED TO THE HOT GASES. IT IS THEREFORE DESIRABLE TO HOLD TOLERANCE ON TAPPED HOLE DEPTH TO AS CLOSE A TOLERANCE AS POSSIBLE WITHOUT UNDUE HARDSHIP IN MACHINING.

ASSEMBLY OF INSERT: IT IS RECOMMENDED THAT INSERT BE ASSEMBLED WITH FLAT TOP OF SERRATED END BELOW THE BOSS SURFACE THE DISTANCE SHOWN ON DRAWING. AFTER ASSEMBLY, INSERT SHOULD BE EXPANDED TO ASSURE PROPER SEATING ESPECIALLY WHEN CLOSE CONCENTRICITY LIMITS ARE DESIRED. THE FLAT SERRATED END OF COIL IS THEN OFF-SET .020-.030, DEPTH OF OFFSET DECREASING UNIFORMLY TO ZERO IN APPROXIMATELY 1/3 TURN, AND STAKED. TANG IS THEN BROKEN OFF AT NOTCH. ASSEMBLY, EXPANDING AND TANG BREAK-OFF TOOLS AND OFF-SET AND STAKING TOOLS ARE STANDARD WITH MANUFACTURER.

REVISED Ⓐ 8 Oct 63

APPROVED 24 Apr 57

THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE ENGINE AND PROPELLER UTILITY PARTS COMMITTEE OF THE SAE.

P.A. USAF-ASD Other Cust Ⓐ Navy-Wepa	TITLE BOSSES, 18-1.50MM SPARK PLUG THREAD HELICAL INSERT, STANDARD DIMENSIONS FOR	MILITARY STANDARD MS9071 (ASG)
PROCUREMENT SPECIFICATION None	SUPERSEDES:	SHEET 1 OF 1

...Department of the Navy, ... Air Force) and the ... and is mandatory for use by that activity. All other military activities are requested to employ this standard where suitable

DD FORM 1 SEP 57 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

PROJECT E85-57-1

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